IEEE BUBT ACADEMIC RESEARCH SHOWCASE 2026 · 13–14 NOVEMBER · TRACK 1: INTELLIGENT COMPUTING, AI & DATA TECHNOLOGIES

How Many Sentinel-1 Scenes Are Enough?

Scene-Count Sensitivity, Agreement-Gated Reliability and Prioritized Review of Post-Monsoon Riverbank Erosion Maps — Jamuna River, Bangladesh

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KEY MESSAGE **2–3 Sentinel-1 scenes give usable erosion maps. Abstention removes false alarms but hides real erosion; triage + prioritized review recovers most of it with 10–20 % review effort. There is no automatic accuracy gain.**

01 ABSTRACT

Sentinel-1 radar maps riverbank erosion through clouds, but a quick post-monsoon map has few comparable scenes: only 2–3 ascending scenes each November over our 12-km Jamuna reach (Kurigram). We reproduced the Freihardt & Frey (2023) threshold rule on Earth Engine σ^0 and scored every subset of 1–8 post-monsoon scenes (121 maps per rule) against an independent Sentinel-2 reference, tuning on 2024 only. With the published thresholds, false erosion fell from 74 ha (1 scene) to 24 ha (3 scenes), and F1 rose from 0.56 to 0.71, reaching 0.82 with 8 scenes. Agreement + abstention (v1) cut false erosion below 3 ha. However, it abstained on 9–12 % of pixels holding 14–21 ha of real erosion, so it did not beat a tuned baseline. A pre-registered v2 sorts pixels into high-confidence / review / stable and ranks review patches using map-date SAR only. On an independent reach (Sirajganj–Kazipur), prioritized review beat random review in 11 of 12 pre-registered tests, often by small margins. v2 adds no automatic accuracy gain.

02 REAL-WORLD PROBLEM

- The Jamuna erodes farmland and homesteads every monsoon — CEGIS monitored ~348 ha of Jamuna bank erosion in 2025.
- Post-monsoon maps of lost land help agencies prepare before the next monsoon.
- Radar sees through clouds, but a quick map has few comparable scenes: **only 2–3 ascending Sentinel-1 scenes per November (2017–2025) over our reach.**
- Speckle, wet sand, flooded fields and lingering floodwater can create false erosion — or hide real erosion.

03 EXISTING RESEARCH → GAP

Study	What it did	Missing for our question
Freihardt & Frey 2023 (NHESS)	Sentinel-1 + GEE threshold erosion maps, Jamuna; window & filter effects on backscatter; land-cover OA 87.5–90.6 %	Erosion-map accuracy vs scene count; erosion checked visually only
Shampa et al. 2025 (Front. Earth Sci.)	Delft3D erosion early warning, Brahmaputra–Jamuna	Forecasting task; no SAR
CEGIS (annual report)	Operational erosion prediction from Landsat	30 m; not a SAR reliability test
MCDWD · RADD · GFM · Li et al. 2026	m-of-k confirmation / abstention in flood & forest products	Not tested for riverbank erosion maps

Gap: to our knowledge, no reviewed study measured erosion-map accuracy across Sentinel-1 scene budgets with agreement + abstention + coverage reporting, against an independent optical reference.

04 RESEARCH QUESTIONS

RQ1 How do Precision, Recall, F1, IoU change as scenes decrease (k = 8 → 1)?

RQ2 Does m-of-k agreement cut false erosion, and at what recall cost?

RQ3 Does abstaining near the threshold raise precision — and how much coverage is lost?

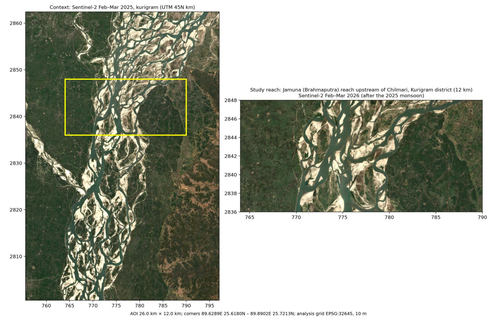
RQ4 Timeliness vs reliability of maps from the first 1–3 scenes after 1 Nov.

RQ5 Can map-date SAR evidence rank possible-erosion patches so a limited review recovers more erosion than random review — on an independent reach?

05 INDEPENDENT REFERENCE & CHECK

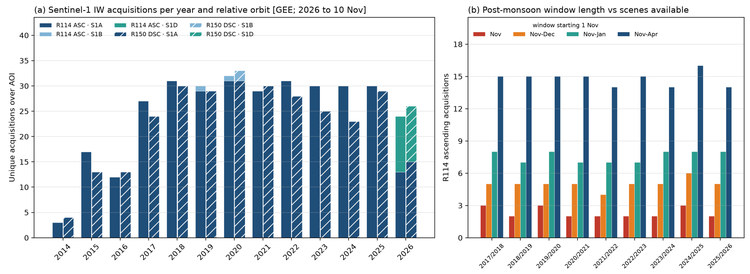
- Sentinel-2 L2A (different sensor): vegetated (NDVI ≥ 0.30) in Feb–Mar before → water (MNDWI ≥ 0) or bare sand (NDVI < 0.20) in the next Feb–Mar, connected to the channel, in a 1.5 km outer-bank strip.
- Visual check of 120 stratified points (blind to label): 36/39 reference-erosion points truly eroded (92 %); 47/47 near-bank non-erosion points correct (100 %). AI-interpreted; team spot-check pending.
- Independent reach (Sirajganj, 2025): 10/15 erosion points (67 %; 15 undecidable), 27/29 non-erosion (93 %). An optical reference — NOT field ground truth.

06 STUDY AREA & DATA



- 12 km Jamuna reach upstream of Chilmari, Kurigram — AOI 26 × 12 km, 25.67°N 89.76°E
- SAR: Sentinel-1 GRD σ^0 (Google Earth Engine COPERNICUS/S1_GRD); VV, IW, orbit 114 ascending, 10 m
- Reference: Sentinel-2 L2A
- Validation monsoon 2024 · test monsoon 2025
- Independent reach (pre-registered): Sirajganj–Kazipur, 28 × 12 km, 24.70°N 89.71°E

07 SENTINEL-1 SCENE AVAILABILITY (FIG. A)



Unique acquisitions over the AOI (Google Earth Engine inventory, cross-checked with ASF and Planetary Computer). Few scenes = orbit geometry and time window, not clouds. Sentinel-1B added only 4 scenes; Sentinel-1D replaced 1A in July 2026.

08 BASELINE & SAGE-RIVER

Sentinel-1 VV · R114 asc
angle 30–45° · circle r=7 (linear) → dB

→ **Pre composite**
mean Nov–Apr (dry season)

→ **Post scenes**
k of 8 (Nov–Jan)

BASELINE
(F&R 2023)
pre > T_pre AND
mean(post) < T_post

A: AGREEMENT
≥ m of k scenes
below T_scene

G: GATING
|value − T| ≤ δ dB
→ UNCERTAIN

SAGE A+G
per-scene ±δ votes
+ m-of-k

v1 output: EROSION · NON-EROSION · UNCERTAIN (abstained, counted in coverage)
v2 triage: HIGH-CONFIDENCE (v1 + bank-connected) · REVIEW (ranked) · LIKELY STABLE

Sentinel-2 reference
veg → sand/water, channel-connected

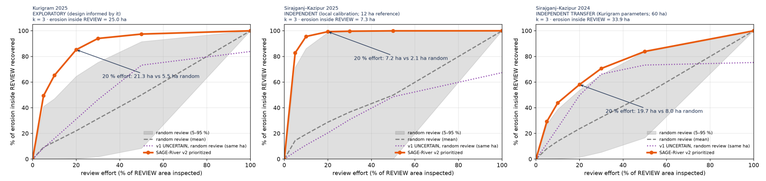
Score on outer-bank strip
P, R, F1, IoU on decided px + coverage

S (season matching) is part of the baseline — a common control, not claimed as new. Gating is an abstention rule, not a calibrated probability.

09 EXPERIMENTAL DESIGN & INTEGRITY

- All tuning on 2024 only (thresholds; m and δ by F1 with coverage ≥ 90 %), frozen with a timestamp, then ONE test on 2025
- Fairness control: a baseline whose thresholds are tuned on the same 2024 erosion F1
- Exp. A: all C(8,k) subsets of the Nov–Jan pool, k = 1, 2, 3, 6, 8 (121 maps per rule)
- Exp. B: first 1, 2, 3 scenes after 1 Nov — no future scenes; same reference & domain for every rule
- v2 design + code frozen (md5, timestamps) before any Sirajganj SAR was processed; Kurigram 2025 v2 = exploratory

10 SAGE-RIVER V2: TRIAGE + PRIORITIZED REVIEW

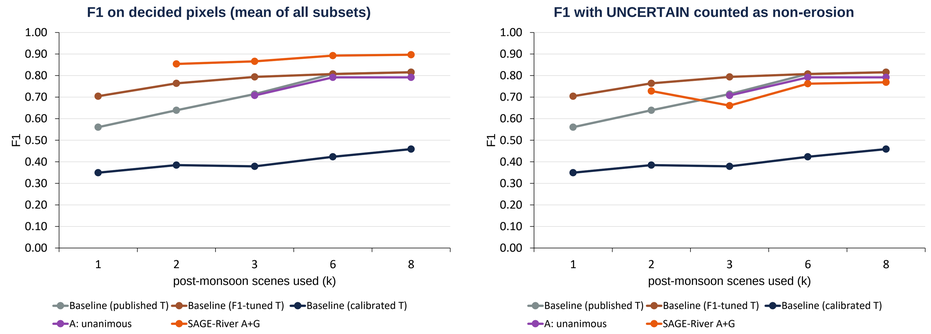


REVIEW = possible erosion within 100 m of the SAR bank, ranked by a 2024-trained logistic model on map-date SAR features only. Review = simulated oracle analyst (upper bound). Random = 500 orders.

k = 3	Status	Tuned F1	v2 HC F1	Recovered @ 20 %: v2 / random / v1-U	Both criteria met
Kurigram 2025	exploratory	0.79	0.66	21.3 / 5.5 / 7.8 ha	k2✓ k3✓ k6✓ k8✓
Sirajganj 2025 (local T)	INDEPENDENT	0.46	0.33	7.2 / 2.1 / 1.5 ha	k2✓ k3✓ k6✓ k8✗
Sirajganj 2024 (Kurigram T)	transfer	0.68	0.54	19.7 / 8.0 / 16.8 ha	k2✓ k3✓ k6✓ k8✓
Sirajganj 2025 (Kurigram T)	transfer	0.53	0.10	9.6 / 2.7 / 3.9 ha	k2✓ k3✓ k6✓ k8✓

Criteria (pre-registered): (a) prioritized > random mean and 95th percentile at 20 %; (b) HC false erosion ≤ v1. HC F1 = full-area, no review. v1-U = random review of v1 UNCERTAIN, same ha. Source: outputs/tables/v2_preregistered_criteria.csv

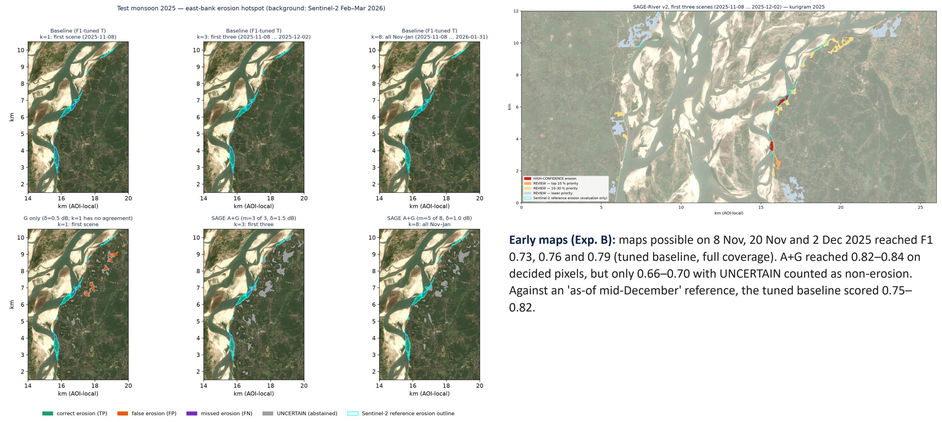
11 RESULTS — TEST MONSOON 2025 (PARAMETERS FROZEN ON 2024)



k	Rule	Precision	Recall	F1	F1 (U = non)	IoU	Coverage	False ha	Missed ha	Abstained ha
1	Baseline (published T)	0.46	0.77	0.56	0.56	0.40	100%	74.4	13.7	0.0
1	Baseline (F1-tuned T)	0.80	0.67	0.70	0.70	0.56	100%	24.3	19.7	0.0
1	Baseline (calibrated T)	0.23	0.78	0.35	0.35	0.21	100%	172.5	12.8	0.0
3	Baseline (published T)	0.67	0.77	0.71	0.71	0.56	100%	24.1	13.3	0.0
3	Baseline (F1-tuned T)	0.88	0.72	0.79	0.79	0.66	100%	6.0	16.2	0.0
3	Baseline (calibrated T)	0.25	0.80	0.38	0.38	0.24	100%	149.0	11.9	0.0
3	A: unanimous	0.71	0.72	0.71	0.71	0.55	100%	19.8	16.3	0.0
3	SAGE-River A+G	0.98	0.77	0.87	0.66	0.76	89%	0.5	8.6	21.0
8	Baseline (published T)	0.86	0.78	0.82	0.82	0.69	100%	7.5	13.1	0.0
8	Baseline (F1-tuned T)	0.86	0.77	0.82	0.82	0.69	100%	7.1	13.5	0.0
8	Baseline (calibrated T)	0.32	0.81	0.46	0.46	0.30	100%	100.2	11.5	0.0
8	A: unanimous	0.96	0.67	0.79	0.79	0.66	100%	1.5	19.3	0.0
8	SAGE-River A+G	0.97	0.83	0.90	0.77	0.81	88%	1.2	7.4	13.9

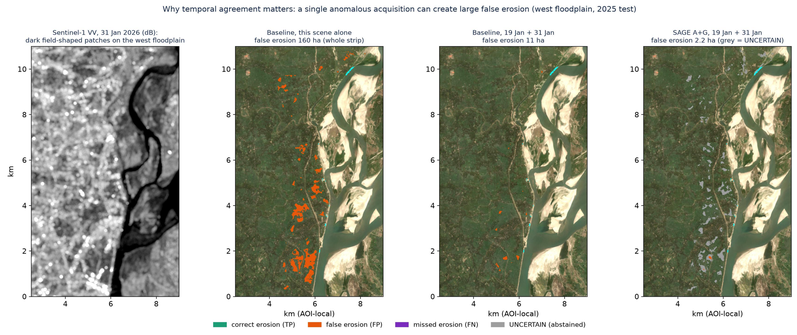
Means over all C(8,k) subsets (overlapping, not independent). Reference erosion = 58.8 ha. Precision/recall on decided pixels; 'F1 (U = non)' counts UNCERTAIN as non-erosion. Source: outputs/tables/experiment_A_summary_gee.csv

12 MAPS (EAST-BANK HOTSPOT) AND EARLY MAPS



Early maps (Exp. B): maps possible on 8 Nov, 20 Nov and 2 Dec 2025 reached F1 0.73, 0.76 and 0.79 (tuned baseline, full coverage). A+G reached 0.82–0.84 on decided pixels, but only 0.66–0.70 with UNCERTAIN counted as non-erosion. Against an 'as-of mid-December' reference, the tuned baseline scored 0.75–0.82.

13 A BAD SCENE, AUDITED: INUNDATED FIELDS — NOT A PROCESSING BUG



Root-cause audit: no processing error. NoData, alignment (0-px shift), radiometry, mosaics and thresholds were all checked. The dark 31 Jan 2026 patches are temporarily inundated fields, a median 584 m from the channel. Evidence: GEE σ^0 –15.0 dB vs –8.2 dB on control fields; also dark in the R150 descending scene; Sentinel-2 shows 88–99 % water from 28 Jan to 12 Feb. The scene was kept in every experiment. Alone it gives 160 ha of false erosion (tuned T); 2-scene A+G maps that contain it average 1.6 ha.

14 KEY FINDINGS · CONTRIBUTIONS · LIMITATIONS

Findings

Scene budget: published-rule false erosion 74 → 24 → 7.5 ha (k = 1 → 3 → 8); tuned-baseline F1 0.70 → 0.79 → 0.82.

v1 A+G: decided-pixel F1 0.85–0.90 with < 3 ha false erosion, but 14–21 ha of real erosion was abstained. Counting UNCERTAIN as non-erosion, it was no better than the tuned baseline (Kurigram and Sirajganj).

v2 at 20 % review (Kurigram k = 3, exploratory): 21.3 ha recovered vs 5.5 ha for random review. Independent Sirajganj: both criteria met in 11/12 tests; failed at k = 8 (local 2025).

No automatic gain: v2 high-confidence F1 = v1, below the tuned baseline in every test. Single-scene maps still fail (31 Jan: 104 ha false).

Contributions

Erosion-map scene-budget benchmark (121 subsets × 7 rules) of the published Sentinel-1/GEE method against an independent Sentinel-2 reference.

Abstention reported with coverage and abstained-erosion area — no hidden pixels.

Three-class triage + map-date review ranking, pre-registered (md5 + timestamps) and tested on an independent reach.

Root-cause audit of a failing scene (inundated fields, three independent sources); open pipeline with all CSVs.

Limitations

Sentinel-2 is an optical reference, not field truth. Sirajganj 2025 has only 12 ha of erosion, with 67 % agreement on erosion points.

The reviewer is simulated as an oracle, and the REVIEW class covers 184–1218 ha. In Sirajganj 2024, randomly reviewing v1 UNCERTAIN pixels was almost as good at 20 % effort.

Kurigram 2025 v2 numbers are exploratory. Only two reaches of one river were tested, and overlapping subsets are not independent samples.

Historical mapping only — no forecasting, no PINN.

REFERENCES [1] Freihardt & Frey (2023) NHESS 23:751–770, doi:10.5194/nhess-23-751-2023 · [2] Freihardt & Frey (2022) code, doi:10.5281/zenodo.7253121 · [3] Shampa et al. (2025) Front. Earth Sci. 13:1570577, doi:10.3389/feart.2025.1570577 · [4] Aishi & Fahim (2024) GNHR 15:2399668, doi:10.1080/19475705.2024.2399668 · [5] Langhorst & Pavelsky (2023) JGR-ES 128:e2022JF006774 · [6] CEGIS (2026) Prediction of Riverbank Erosion 2026, cegisbd.com/news/3011 · [7] Reiche et al. (2021) ERL 16:024005 · [8] Krullikowski et al. (2023) IEEE JSTARS 16:6917 · [9] Li, Matgen & Chini (2026) IEEE TGRS 64:4202520 · [10] Peña-Luque et al. (2021) Remote Sens. 13:3279 · Data: Copernicus Sentinel-1/2 (ESA) via Google Earth Engine, Microsoft Planetary Computer, ASF.

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Code, data & CSVs: SAGE-River/ (README)

Sentinel-2 is an optical reference, not field ground truth.